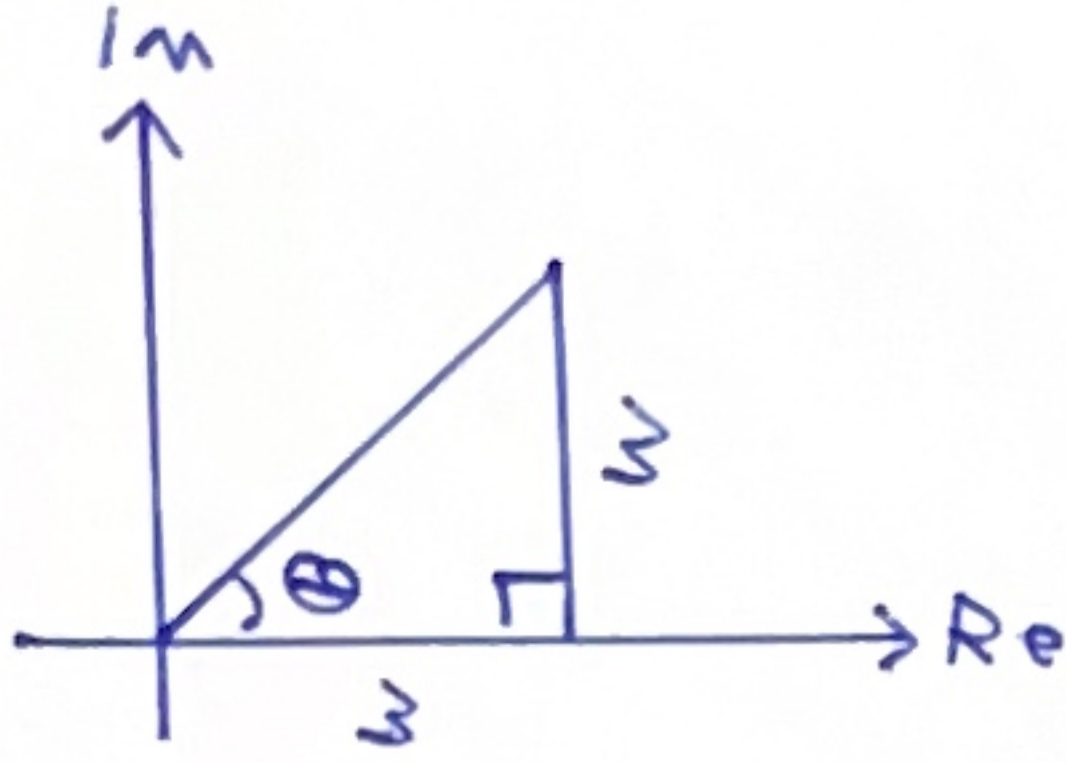


ANSWER SCHEME
PSPM 2015/2016

Q#	ANSWER	MARKS
Q1	$4^{y-2} = \frac{1}{3^{1-y}}$ $4^{y-2} = 3^y \text{ — (1)}$ $\log 4^{y-2} = \log 3^y \text{ — (1)}$ $(y-2) \log 4 = y \log 3 \text{ — (1)}$ $y \log 4 - 2 \log 4 = y \log 3 \text{ — (1)}$ $y \log 4 - y \log 3 = 2 \log 4$ $y (\log 4 - \log 3) = 2 \log 4 \text{ — (1)}$ $y = \frac{2 \log 4}{\log 4 - \log 3} = 9.638 \text{ — (1)}$	Q1 = 6 marks
Q2	$2 + \log_2 x = 15 \log_x 2$ $2 + \log_2 x = 15 \left[\frac{\log_2 2}{\log_2 x} \right] \text{ — (1)}$ $2 + \log_2 x = \frac{15}{\log_2 x}$ $5x + 4 = \log_2 x,$ $5x + 4 = \frac{15}{x}$ $5x^2 + 24 - 15 = 0 \text{ — (1)}$ $(x-3)(x+5) = 0 \text{ — (1)}$ $x = 3 \text{ OR } x = -5 \text{ — (1)}$ $\log_2 x = 3 \quad \log_2 x = -5 \text{ — (1)}$ $x = 8 \text{ — (1)} \quad x = \frac{1}{32} \text{ — (1)}$	Q2 = 7 marks

Q#	ANSWER	REMARKS
Q3 (a)	$\bar{z}_1 = 3 + 3i$  $ \bar{z}_1 = \sqrt{3^2 + 3^2} \text{ --- (1)}$ $= 3\sqrt{2} \text{ --- (2)}$ $\theta = \tan^{-1} \frac{3}{3}$ $= \frac{\pi}{4} \text{ --- (3)}$ $\therefore \bar{z}_1 = 3\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \text{ --- (4)}$	_____ Q3(a) = 4 marks
(b)	$\frac{\bar{z}_1 z_2}{13} = \frac{(3 + 3i)(3 + 2i)}{13} \text{ --- (5) sub with correct } \bar{z}_1$ $= \frac{9 + 6i + 9i + 6i^2}{13} \text{ --- (6)}$ $= \frac{9 + 15i + 6(-1)}{13}$ $= \frac{3 + 15i}{13}$ $= \frac{3}{13} + \frac{15i}{13} \text{ --- (7)}$	

Q#	ANSWER	REMARKS
	$\frac{1^2}{-z_2} = \frac{1^2 \cdot 1}{-(3+2i)}$ $= \frac{-1}{-3-2i} \cdot \frac{-3+2i}{-3+2i} \text{ — (I)}$ $= \frac{3i - 2i^2}{9 - 4i^2}$ $= \frac{3i - 2(-1)}{9 - 4(-1)} \text{ — (II)}$ $= \frac{2 + 3i}{13}$ $= \frac{2}{13} + \frac{3}{13}i$ $\overline{\frac{1^2}{-z_2}} = \frac{2}{13} - \frac{3}{13}i \text{ — (III)}$ $\therefore \frac{\overline{z_1} z_2}{13} + \left[\frac{1^2}{-z_2} \right]$ $= \frac{5}{13} + \frac{12}{13}i + \frac{2}{13} - \frac{3}{13}i \text{ — (IV)}$ $= \frac{7}{13} + \frac{9}{13}i \text{ — (V)}$	_____ Q3(b) = 8 MARKS

Q#	ANSWER	REMARKS
Q4 (a)	$\text{Let } f(x_1) = f(x_2)$ $\sqrt{3-2x_1} = \sqrt{3-2x_2}$ $3-2x_1 = 3-2x_2$ $2x_1 = 2x_2$ $x_1 = x_2 \quad \text{--- (1)}$ $f(x)$ is a one to one function because $x_1 = x_2$. $\therefore$ shown! --- (2)	Q4 (a) = 2 MARKS
(b)	$\left. \begin{aligned} \Delta f &= 3-2x \geq 0 \\ 2x &\leq 3 \\ x &\leq \frac{3}{2} \end{aligned} \right\} \text{--- (1)}$ $R_f = [0, \infty) \quad \text{--- (2)}$ $\Delta f = (-\infty, \frac{3}{2}] \quad \text{--- (3)}$	Q4 (b) = 3 MARKS
(c)	$f f^{-1}(x) = x$ $\sqrt{3-2f^{-1}(x)} = x \quad \text{--- (1)}$ $3-2f^{-1}(x) = x^2$ $2f^{-1}(x) = 3-x^2$ $f^{-1}(x) = \frac{3-x^2}{2} \quad \text{--- (2)}$ $\Delta f^{-1} = R_f = [0, \infty) \quad \text{--- (3)}$ $R_{f^{-1}} = \Delta f = (-\infty, \frac{3}{2}] \quad \text{--- (4)}$	Q4 (c) = 4 MARKS

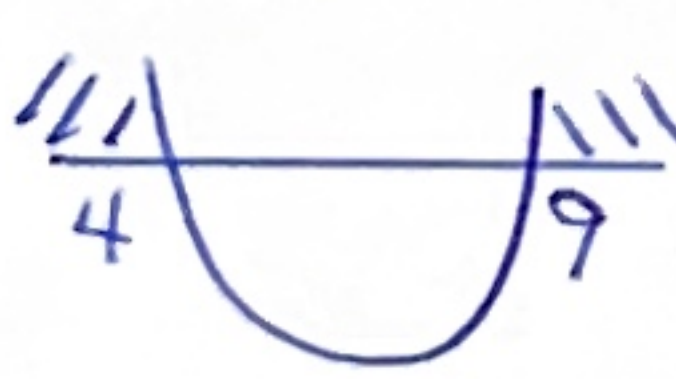
Q#	ANSWER	REMARKS
(Q)	$y = f(x)$ $y = f^{-1}(x)$ $y = x$	$\textcircled{R1} \ y = f(x)$ $\textcircled{R1} \ y = f^{-1}(x)$ $\textcircled{R1} \ \text{all correct}$ Q4(c) = 3 MARKS
Q5 (Q)	$f[f(x)] = x$ $a \left[\frac{ax+2}{3x-4} \right] + 2 = x \quad \text{---} \quad \textcircled{R1}$ $\frac{3 \left[\frac{ax+2}{3x-4} \right] - 4}{a^2x + 2a + 2(3x-4)} = x \quad \text{---} \quad \textcircled{R2}$ $\frac{3ax + 6 - 4(3x-4)}{3x-4} = x \quad \text{---} \quad \textcircled{R3}$ $\frac{a^2x + 2a + 6x - 8}{3ax + 6 - 12x + 16} = x \quad \text{---} \quad \textcircled{R4}$ $(a^2+6)x + 2a - 8 = 3ax^2 - 12x^2 + 22x \quad \text{---} \quad \textcircled{R5}$ COMPARE Coef. of x ;	
	$a + 6 = 22$	

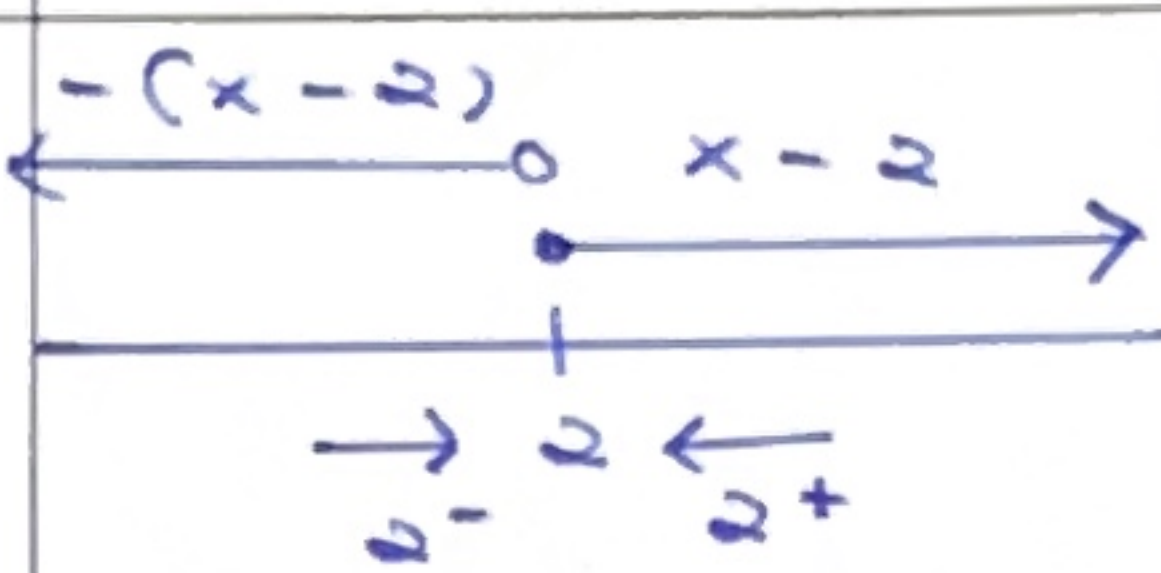
$$a^2 = 16$$

$$a = \pm 4$$

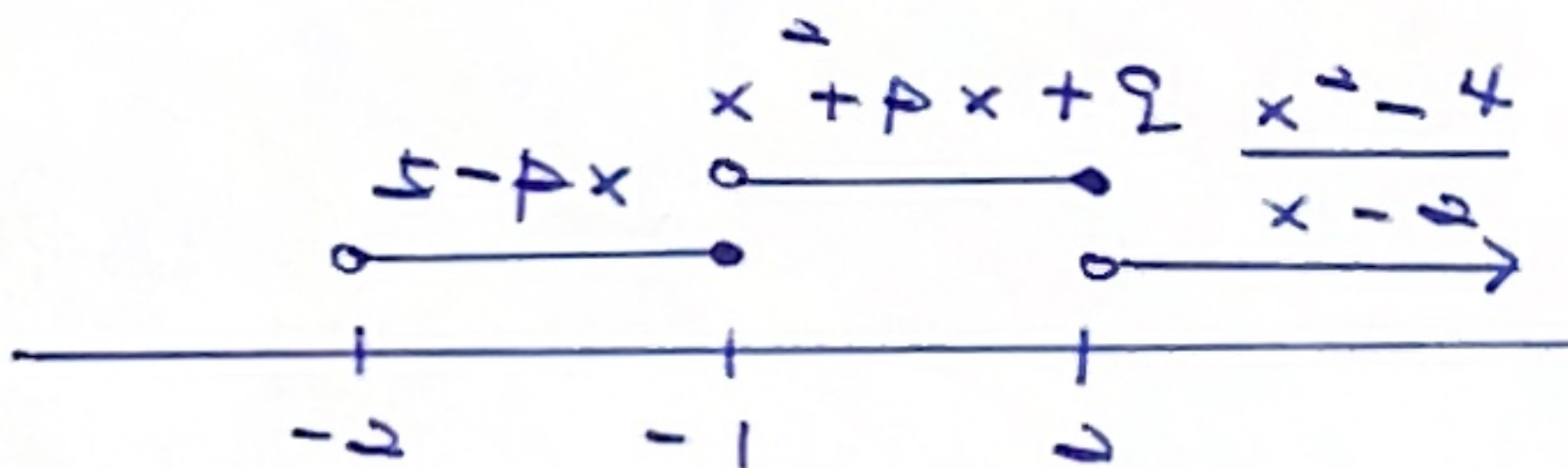
Q#	ANSWER	REMARKS
	Compare coef of x^0 ; $2a - 8 = 0 \quad \text{--- (K)}$ $2a = 8$ $a = 4$ $\therefore a = 4 \quad \text{--- (J)}$ (b) $g \circ f(x) = e^{-\ln 3x+2 +2} \quad \text{--- (K)}$ $= e^{\ln(3x+2)^{-1}+2}$ $= \frac{1}{3x+2} + 2 \quad \text{--- (K)}$ $= \frac{1+2(3x+2)}{3x+2}$ $= \frac{6x+5}{3x+2} \quad \text{--- (J)}$ Let $g \circ f(x) = h(x)$ $h[h^{-1}(x)] = x$ $\frac{6h^{-1}(x)+5}{3h^{-1}(x)+2} = x \quad \text{--- (K)}$ $6h^{-1}(x)+5 = 3xh^{-1}(x)+2x$ $6h^{-1}(x)-3xh^{-1}(x) = 2x-5$ $h^{-1}(x) = \frac{2x-5}{6-3x} = (g \circ f)^{-1}(x) \quad \text{--- (J)}$ $\therefore (g \circ f)^{-1}(3) = \frac{2(3)-5}{6-3(3)} \quad \text{--- (K)}$	$Q5(a) = 6 \text{ MARKS}$ $Q5(b) = 7 \text{ MARKS}$
	$= -\frac{1}{3} \quad \text{--- (J)}$	

Q#	ANSWER	REMARKS																																								
Q6 (4)	$\left \frac{x-1}{x+3} \right > 2$ $\frac{x-1}{x+3} > 2 \quad \text{OR} \quad -\left(\frac{x-1}{x+3}\right) > 2 \quad \text{--- (K1)}$ $\frac{x-1}{x+3} - 2 > 0$$\frac{x-1-2(x+3)}{x+3} > 0$$\frac{-x-7}{x+3} > 0$$\text{CN : } x < -7, -3$<table><tr><td>Interval</td><td>$(-\infty, -7)$</td><td>$(-7, -3)$</td><td>$(-3, \infty)$</td></tr><tr><td>Value x</td><td>-8</td><td>-5</td><td>0</td></tr><tr><td>$-x-7$</td><td>+</td><td>-</td><td>-</td></tr><tr><td>$x+3$</td><td>-</td><td>-</td><td>+</td></tr><tr><td>$\frac{-x-7}{x+3}$</td><td>-</td><td>(+)</td><td>-</td></tr></table>$\therefore (-7, -3)$ $\frac{x-1}{x+3} + 2 < 0$$\frac{x-1+2(x+3)}{x+3} < 0$$\frac{3x+5}{x+3} < 0 \quad \text{--- (K2)}$$\text{CN : } x < -\frac{5}{3}, -3$<table><tr><td>Interval</td><td>$(-\infty, -3)$</td><td>$(-3, -\frac{5}{3})$</td><td>$(-\frac{5}{3}, \infty)$</td></tr><tr><td>Value x</td><td>-4</td><td>-2</td><td>0</td></tr><tr><td>$3x+5$</td><td>-</td><td>-</td><td>+</td></tr><tr><td>$x+3$</td><td>-</td><td>+</td><td>+</td></tr><tr><td>$\frac{3x+5}{x+3}$</td><td>+</td><td>(-)</td><td>+</td></tr></table>$\therefore (-3, -\frac{5}{3}) \quad \text{--- (J1)}$ $\therefore (-7, -3) \cup (-3, -\frac{5}{3}) \quad \text{--- (J1)}$ $\text{Q6(4)} = 6 \text{ marks}$	Interval	$(-\infty, -7)$	$(-7, -3)$	$(-3, \infty)$	Value x	-8	-5	0	$-x-7$	+	-	-	$x+3$	-	-	+	$\frac{-x-7}{x+3}$	-	(+)	-	Interval	$(-\infty, -3)$	$(-3, -\frac{5}{3})$	$(-\frac{5}{3}, \infty)$	Value x	-4	-2	0	$3x+5$	-	-	+	$x+3$	-	+	+	$\frac{3x+5}{x+3}$	+	(-)	+	
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$x+3$	-	+	+																																							
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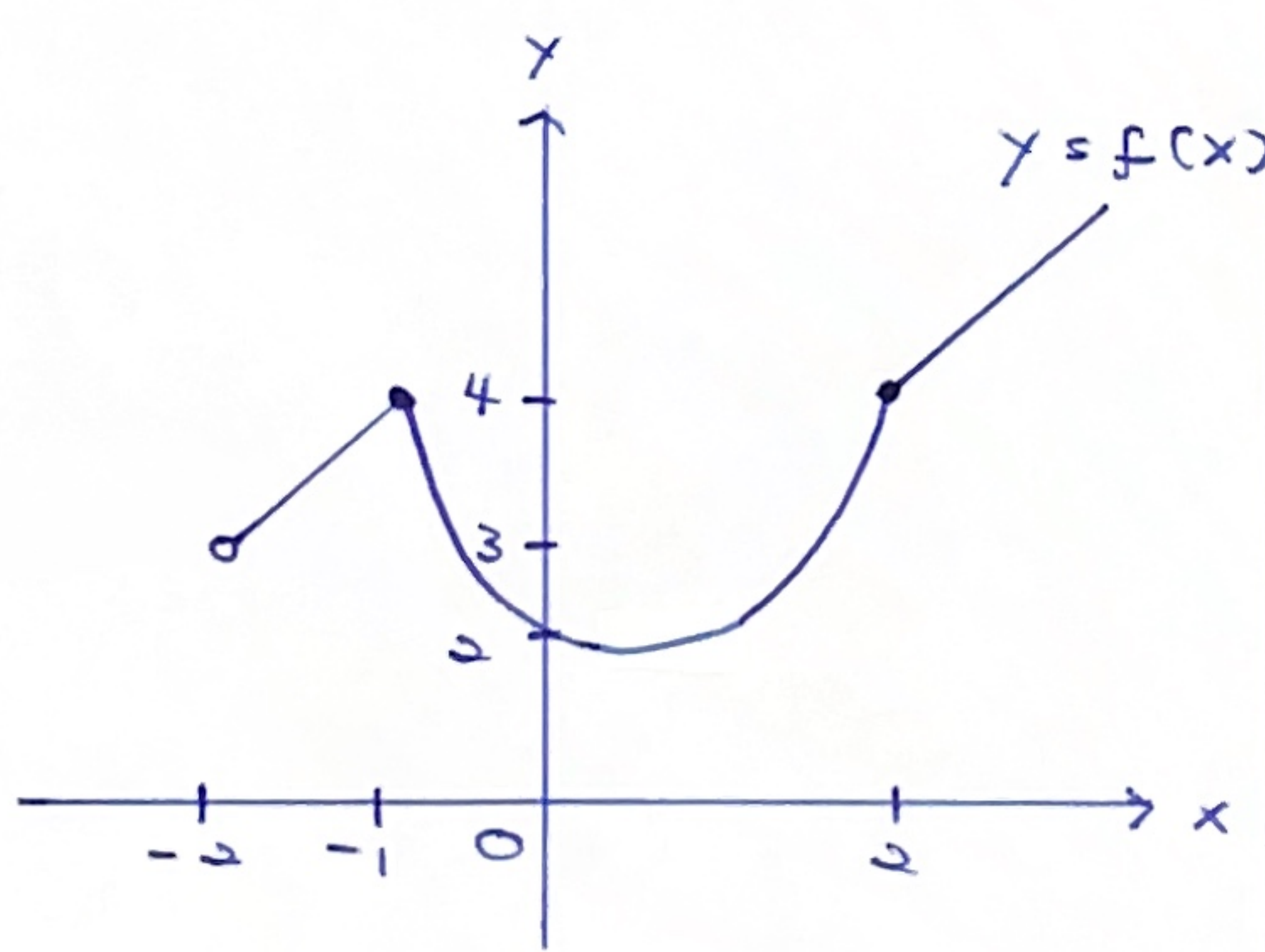
Q#	ANSWER	REMARKS
(b)	$\frac{2^x \times 4^{2x}}{8^x}$ $= \frac{2^x \cdot 2^{4x}}{2^{3x}} \text{ — (Fi)}$ $= 2^{x+4x-3x} \text{ — (Fi)}$ $= 2^{2x} \therefore \text{shown!} \text{ — (Ji)}$ $\frac{2^x \cdot 4^{2x}}{8^x} - 13(2^x) + 36 \geq 0$ $2^{2x} - 13(2^x) + 36 \geq 0 \text{ — (Bi)}$ Let $y = 2^x$, $y^2 - 13y + 36 \geq 0$ $(y-9)(y-4) \geq 0 \text{ — (Fi)}$ on: $y \leq 9, y \leq 4$  $y \leq 4 \text{ OR } y \geq 9$ $2^x \leq 4 \qquad 2^x \geq 9 \text{ — (Ji)}$ $x \leq \log_2 4 \qquad x \geq \log_2 9 \text{ — (Fi)}$ $x \leq 2 \qquad x \geq 3.17$ $\therefore (-\infty, 2] \cup [3.17, \infty) \text{ — (Ji)}$	_____ Q6(b) = 9 marks

Q#	ANSWER	REMARKS
Q7 (a)	$ x-2 = \begin{cases} x-2, & x \geq 2 \\ -(x-2), & x < 2 \end{cases}$ $\lim_{x \rightarrow 2^-} \frac{x^2 + 4x - 12}{-(x-2)} \text{ --- (K)}$ $\therefore \lim_{x \rightarrow 2^-} \frac{(x-2)(x+6)}{-(x-2)} \text{ --- (K)}$ $\therefore \lim_{x \rightarrow 2^-} -(x+6)$ $\therefore -(2+6)$ $\therefore -8$ $\lim_{x \rightarrow 2^+} \frac{x^2 + 4x - 12}{x-2}$ $\therefore \lim_{x \rightarrow 2^+} \frac{(x-2)(x+6)}{x-2}$ $\therefore \lim_{x \rightarrow 2^+} (x+6)$ $\therefore 2+6$ $\therefore 8 \text{ --- (J) both } \lim_{x \rightarrow 2^-} f(x) \text{ and } \lim_{x \rightarrow 2^+} f(x)$ Since $\lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$, $\therefore \lim_{x \rightarrow 2} \frac{x^2 + 4x - 12}{ x-2 } \text{ does not exist. --- (J)}$	
		Q7(a) = 4 MARKS

Q#	ANSWER	REMARKS
Q7 (b)	$\lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{1 - x} \cdot \frac{1 + \sqrt{x}}{1 + \sqrt{x}} \text{ — (T)}$ $= \lim_{x \rightarrow 1} \frac{1 - x}{(1 - x)(1 + \sqrt{x})}$ $= \lim_{x \rightarrow 1} \frac{1}{1 + \sqrt{x}} \text{ — (T)}$ $= \frac{1}{1 + \sqrt{1}}$ $= \frac{1}{2} \text{ — (J)}$	Q7 (b) = 3 MARKS
Q8 (a)	$f(x) = \cot(4x^2 + 1)^{\frac{1}{2}}$ $f'(x) = -\cot \sec^2(4x^2 + 1)^{\frac{1}{2}} \cdot \frac{1}{2}(4x^2 + 1)^{-\frac{1}{2}} \cdot 8x$ $= \frac{-4 \cot \sec^2 \sqrt{4x^2 + 1}}{\sqrt{4x^2 + 1}} \text{ — (T)}$	Q8 (a) = 3 MARKS
Q8 (b)	$u = e^{2x} \quad v = \ln(3x + 4)$ $u' = 2e^{2x} \quad v' = \frac{3}{3x + 4} \text{ — (T)}$ $f'(x) = \ln(3x + 4) \cdot 2e^{2x} + e^{2x} \cdot \frac{3}{3x + 4}$ $= e^{2x} \left[2 \ln(3x + 4) + \frac{3}{3x + 4} \right] \text{ — (J)}$	Q8 (b) = 4 MARKS

Q#	ANSWER	REMARKS
Q9 (a)	$\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2 + 3x}}{5x + 1}$ $= \lim_{x \rightarrow -\infty} \frac{\sqrt{\frac{2x^2}{x^2} + \frac{3x}{x^2}}}{\frac{5x}{x} + \frac{1}{x}} \quad \text{--- (1)}$ $= \lim_{x \rightarrow -\infty} \frac{\sqrt{2 + \frac{3}{x}}}{5 + \frac{1}{x}} \quad \text{--- (2)}$ $= \frac{-\sqrt{2 - 0}}{5 - 0} \quad \text{--- (3)}$ $= -\frac{\sqrt{2}}{5} \quad \text{--- (4)}$	
(b)		
(c)	$\text{At } x = -1,$ $\lim_{x \rightarrow -1^-} (5 - px) = \lim_{x \rightarrow -1^+} (x^2 + px + 2) \quad \text{--- (1)}$ $5 - p(-1) = (-1)^2 + p(-1) + 2$ $5 + p = 1 - p + 2$ $2p - 2 = -4 \quad \text{--- (2)}$	

Q9(a) = 4 MARKS

Q#	ANSWER	REMARKS
	At $x = 2$, $\lim_{x \rightarrow 2^-} (x^2 + px + q) = \lim_{x \rightarrow 2^+} \frac{x^2 - 4}{x - 2} \quad \text{--- (K)}$ $(2^2) + p(2) + q = \lim_{x \rightarrow 2^+} \frac{(x-2)(x+2)}{x-2}$ $4 + 2p + q = 2 + 2$ $2p + q = 0 \quad \text{--- (2)} \quad \text{--- (K) both correct equations}$ (1) + (2) : $\text{--- (K) solve equations}$ $4p = -4$ $p = -1 \quad \text{--- sub into (2)}$ $2(-1) + q = 0$ $q = 2$ $\therefore p = -1$ and $q = 2 \quad \text{--- (J) (J)}$	
(ii)		When $x = -2$, $y = 5 + (-2) = 3$ When $x = -1$, $y = 5 + (-1) = 4$ --- (K) one piece correct --- (K) 2 pieces correct --- (K) all correct with labels

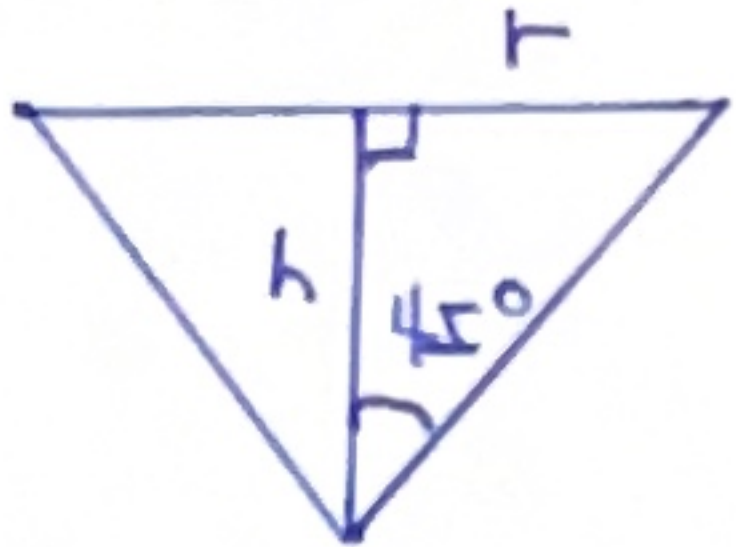
Q9(b)(ii) = 3 MARKS

Q#	ANSWER	REMARKS
Q10 (a)	$\frac{dx}{dt} = 1 + \frac{1}{t^2} = \frac{t^2 + 1}{t^2} \text{ --- (F)}$ $\frac{dy}{dt} = 1 - \frac{1}{t^2} = \frac{t^2 - 1}{t^2} \text{ --- (F)}$ $\frac{dy}{dx} = \frac{t^2 - 1}{t^2} \cdot \frac{t^2}{t^2 + 1} \text{ --- (F)}$ $= \frac{t^2 - 1}{t^2 + 1} \text{ --- (J)}$ $\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx}$ $u = t^2 - 1 \quad v = t^2 + 1$ $u' = 2t \quad v' = 2t$ $= \frac{(t^2 + 1)(2t) - (t^2 - 1)(2t)}{(t^2 + 1)^2} \cdot \frac{t^2}{t^2 + 1} \text{ --- (F) } \frac{d}{dt} \left(\frac{dy}{dx} \right)$ $= \frac{2t^3 + 2t - 2t^3 + 2t}{(t^2 + 1)^2} \cdot \frac{t^2}{t^2 + 1} \text{ --- (F)}$ $= \frac{4t}{(t^2 + 1)^3} \text{ --- (J)}$	multiply $\frac{dt}{dx}$ Q10(a) = 7 MARKS

Q#	ANSWER	REMARKS
(b)	At stationary points, $\frac{dy}{dx} = 0$ $\frac{t^2 - 1}{t^2 + 1} = 0 \quad \text{--- (Fi)}$ $t^2 - 1 = 0$ $t^2 = 1$ $t = \pm 1 \quad \text{--- (Ji)}$ When $t = 1$, $x = 1 - \frac{1}{1} = 0$ $y = 1 + \frac{1}{1} = 2$ $\left. \begin{array}{l} x = 1 - \frac{1}{1} = 0 \\ y = 1 + \frac{1}{1} = 2 \end{array} \right\} \text{--- (Fi)}$ When $t = -1$, $x = -1 - \frac{1}{-1} = 0$ $y = -1 + \frac{1}{-1} = -2$ $(0, 2)$ and $(0, -2)$ are the stationary points. --- (Ji) When $t = 1$, $\frac{d^2y}{dx^2} = \frac{4(1)^3}{(1^2 + 1)^3} = \frac{1}{2} (> 0) \quad \text{--- (Fi)}$ When $t = -1$, $\frac{d^2y}{dx^2} = \frac{4(-1)^3}{[(-1)^2 + 1]^3} = -\frac{1}{2} (< 0)$	Q10(b) = 6 MARKS

$\therefore (0, 2)$ is a minimum point --- (Ji)
and $(0, -2)$ is a maximum point.

Q#	ANSWER	REMARKS
Q11 (2)	$u = 2y$ $u' = 2 \frac{dy}{dx}$ $v = (1+x^2)^{\frac{1}{2}}$ $v' = \frac{x}{\sqrt{1+x^2}}$ $2y \frac{dy}{dx} - \left[2y \cdot \frac{x}{\sqrt{1+x^2}} + \sqrt{1+x^2} \cdot 2 \frac{dy}{dx} \right] + 2x = 0$ $2y \frac{dy}{dx} - \frac{2xy}{\sqrt{1+x^2}} - 2\sqrt{1+x^2} \frac{dy}{dx} + 2x = 0$ $2y \frac{dy}{dx} - 2\sqrt{1+x^2} \frac{dy}{dx} = \frac{2xy}{\sqrt{1+x^2}} - 2x$ $\frac{dy}{dx} (2y - 2\sqrt{1+x^2}) = \frac{2xy}{\sqrt{1+x^2}} - 2x$ $\frac{dy}{dx} = \frac{\frac{2xy}{\sqrt{1+x^2}} - 2x}{2y - 2\sqrt{1+x^2}}$ $= \frac{xy - x\sqrt{1+x^2}}{(y - \sqrt{1+x^2})(\sqrt{1+x^2})}$ $= \frac{x(y - \sqrt{1+x^2})}{(y - \sqrt{1+x^2})(\sqrt{1+x^2})}$ $= \frac{x}{\sqrt{1+x^2}}$ shown!	(K) — (K) 2 correct (K) 3 correct (K) attempt to solve (K) — (J) — Q11 correct marks

Q#	ANSWER	REMARKS
(b)	 $\frac{dV}{dt} = 36\pi \text{ cm}^3 \text{ s}^{-1}$	
(c)	$\frac{dh}{dt} = ?$ when $h = 3 \text{ cm}$ $V = \frac{1}{3} \pi r^2 h$ — (1) $\tan 45^\circ = \frac{r}{h}$ $r = h$ — sub into (1) $V = \frac{1}{3} \pi h^3$ — (ii) $\frac{dV}{dh} = \pi h^2$ $\frac{dh}{dt} = \frac{dh}{dV} \cdot \frac{dV}{dt}$ $= \frac{1}{\pi h^2} \cdot 36\pi$ — (iii) $= \frac{36}{h^2}$ — (iv) When $h = 3 \text{ cm}$ $\frac{dh}{dt} = \frac{36}{3^2}$ $= 4 \text{ cm s}^{-1}$ — (v)	_____ Q11 (b) (c) = 4 MARKS

Q#	ANSWER	REMARKS
c)ii)	$V = \frac{1}{3} \pi h^3$ when $h = 18 \text{ cm}$, $V = \frac{1}{3} \pi (18)^3 \text{ — (H)}$ $= 1944 \pi$ $\frac{dV}{dt} = 36 \pi$ $\frac{1944 \pi}{t} = 36 \pi \text{ — (H)}$ $t = 54 \text{ s — (J)}$	Q11(b), (ii) = 3 MARKS

Q#	ANSWER	REMARKS
Q12 (a)	$x^3 + kx^2y - 2y^2 = 0$ $u = x^2 \quad v = y$ $u' = 2x \quad v' = \frac{dy}{dx} \text{ — (M)}$ $3x^2 + k \left[x^2 \frac{dy}{dx} + 2xy \right] - 4y \frac{dy}{dx} = 0 \text{ — (M) (M)}$ At point $(-1, 1)$; $\frac{dy}{dx} = -3$ $3(-1)^2 + k \left[(-1)^2(-3) + 2(-1)(1) \right] - 4(1)(-3) = 0 \text{ — (M)}$ $3 - 5k + 12 = 0$ $5k = 15$ $k = 3 \text{ — (J)}$	Q12(a) = 5 MARKS
(b) (i)	$u = \sinh x \quad v = 1 + \cosh x$ $u' = \cosh x \quad v' = \sinh x \text{ — (M)}$ $\frac{dy}{dx} = \frac{(1 + \cosh x)(\cosh x) - \sinh x(-\sinh x)}{(1 + \cosh x)^2}$ $= \frac{\cosh x + \cosh^2 x + \sinh^2 x}{(1 + \cosh x)^2}$ $= \frac{1 + \cosh x}{(1 + \cosh x)^2}$ $= \frac{1}{1 + \cosh x} \text{ — (J)}$	

Q#	ANSWER	REMARKS
	$u = 1 \quad v = 1 + \cos x$ $u' = 0 \quad v' = -\sin x$ $\frac{d^2 y}{dx^2} = \frac{(1 + \cos x)(0) - 1(-\sin x)}{(1 + \cos x)^2} \quad \text{--- (M)}$ $= \frac{\sin x}{(1 + \cos x)^2} \quad \text{--- (J)}$	_____ Q12(b)(i) = 5 MARKS
(ii)	$\frac{d^2 y}{dx^2} = \frac{\sin x}{1 + \cos x} \cdot \frac{1}{1 + \cos x} \quad \text{--- (M)}$ $= y \frac{dy}{dx} \quad \text{--- (J)}$ $u = y \quad v = \frac{dy}{dx}$ $u' = \frac{dy}{dx} \quad v' = \frac{d^2 y}{dx^2} \quad \text{--- (M)}$ $\frac{d^3 y}{dx^3} = y \frac{d^2 y}{dx^2} + \frac{dy}{dx} \cdot \frac{dy}{dx} \quad \text{--- (M)}$ $\therefore \frac{d^3 y}{dx^3} - y \frac{d^2 y}{dx^2} - \left(\frac{dy}{dx}\right)^2 = 0 \quad \text{--- (J)}$ shown!	_____ Q12(b)(ii) = 5 MARKS